Time Series

Prove ed esami con soluzioni

Parte I: sei mock preesistenti e la prova indicata come ufficiale del 4 giugno 2026, con le rispettive soluzioni.

Parte II: tredici prove generate e tre ulteriori prove generate, tutte con le soluzioni disponibili.

Il carattere ufficiale della prova del 4 giugno e' riportato dal fascicolo preesistente. I mock sono esercitazioni, non appelli ufficiali.

Parte 1

Mock preesistenti e prova ufficiale

I testi sono seguiti immediatamente dalle soluzioni della stessa prova.

Prove 1-6: mock preesistenti. Prova 7: indicata come ufficiale del 4 giugno 2026.

Exam 1 – Mock Exam 1

1. Univariate time series. Consider a time series $\{y_t\}$ generated by the following $ARMA(1,2)$ process:

$$y_t = \beta y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2},$$

where ε_t is a white noise process with variance σ^2, and $\beta, \alpha_1, \alpha_2 \neq 0$.

1. Under what conditions is $\{y_t\}$ stationary? Under stationarity, find $\mathbb{E}(y_t)$.
2. Is it possible that the above process displays the same autocorrelation function of $y_t = \varepsilon_t + \delta \varepsilon_{t-1}$, where $\delta \neq 0$? If so, under what conditions?
3. Under stationarity, given a sample of T observations, find the optimal predictors for $\{y_t\}$ at horizons $h = 1, 2, 3$. Show what happens to $\hat{y}_{T+h|T}$ when $h \to \infty$.

2. Multivariate time series. Consider

$$y_t^* = \alpha + \beta x_t + \gamma x_{t-1} + \delta y_{t-1} + \varepsilon_t,$$

where y_t^* is the desired target value of y, $0 < \delta < 1$, $\{x_t\}$ is stationary with $\mathbb{E}(x_t) = \mu$, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$. The observable value satisfies

$$y_t - y_{t-1} = \lambda(y_t^* - y_{t-1}), \qquad 0 < \lambda < 1.$$

1. Derive an autoregressive distributed lag model for y_t.
2. Find the impact multiplier of x_t on y_t. Is it affected by λ?
3. Find $\mathbb{E}(y_t)$ and the long-run multiplier of a permanent change in x. Explain the role of δ.

Solutions to Exam 1 – Mock Exam 1

1. Write $(1-\beta L)y_t = (1+\alpha_1 L+\alpha_2 L^2)\varepsilon_t$.

1. Stationarity requires the AR root to be outside the unit circle, hence $|\beta|<1$. Since there is no constant, $\mathbb{E}(y_t)=0$.

2. It has the same ACF as an $MA(1)$ if the AR factor cancels with one factor of the MA polynomial:
$$1+\alpha_1 L+\alpha_2 L^2 = (1-\beta L)(1+\delta L).$$
Thus $\alpha_1 = \delta-\beta$ and $\alpha_2 = -\beta\delta$; in canonical invertible form also $|\delta|<1$.

3. With invertibility, the past innovations are known from the sample:
$$\hat{y}_{T+1|T} = \beta y_T + \alpha_1\varepsilon_T + \alpha_2\varepsilon_{T-1},$$
$$\hat{y}_{T+2|T} = \beta\hat{y}_{T+1|T} + \alpha_2\varepsilon_T, \qquad \hat{y}_{T+3|T} = \beta\hat{y}_{T+2|T}.$$
Since $|\beta|<1$, $\hat{y}_{T+h|T}\to 0 = \mathbb{E}(y_t)$.

2.

1. Substitution into the adjustment equation gives
$$y_t = \lambda\alpha + \lambda\beta x_t + \lambda\gamma x_{t-1} + [1-\lambda+\lambda\delta]y_{t-1} + \lambda\varepsilon_t.$$

2. The impact multiplier is $\lambda\beta$. It is affected by λ because only the fraction λ of the target gap is closed within the period.

3. Taking expectations,
$$\mathbb{E}(y_t) = \frac{\alpha+(\beta+\gamma)\mu}{1-\delta}.$$
The long-run multiplier is $(\beta+\gamma)/(1-\delta)$. Larger δ increases persistence in the target equation and therefore amplifies the long-run effect.

Exam 2 – Mock Exam 2

1. Univariate time series. Consider

$$y_t = \kappa + \theta y_{t-1} + \varepsilon_t, \qquad \varepsilon_t = v_t \sqrt{\alpha + \beta \varepsilon_{t-1}^2},$$

where v_t is white noise with $\sigma_v^2 = 1$, independent of ε_{t-1}.

1. What assumptions on θ, α, β ensure weak stationarity of $\{y_t\}$?
2. Find the conditional and unconditional mean and variance of $\{y_t\}$.
3. Given a sample of length T, find forecasts for y_{T+h} and for $\sigma_{T+h}^2 = \mathbb{E}(\varepsilon_{T+h}^2 \mid \mathcal{I}_T)$ at $h = 1, 2, 3$. What happens as $h \to \infty$?

2. Multivariate time series. Consider

$$y_t = \delta + \beta_1 x_{t+1}^* + \beta_2 x_t + \varepsilon_t,$$

where x_{t+1}^* is the expectation of x_{t+1} formed at time t, and

$$x_{t+1}^* - x_t^* = \lambda(x_t - x_t^*), \qquad 0 < \lambda < 1.$$

Assume $\{x_t\}$ is stationary and $\varepsilon_t \sim \mathrm{WN}(0, \sigma^2)$.

1. Derive an ADL model for y_t.
2. Find the impact multiplier, the interim multiplier after one period, and the long-run multiplier. Comment on the effect of λ.
3. Construct the equilibrium-correction representation for Δy_t.

Solutions to Exam 2 – Mock Exam 2

1.

1. Weak stationarity requires $|\theta| < 1$, $\alpha > 0$, $\beta \geq 0$, and $\beta < 1$. The last condition gives finite $\mathbb{E}(\varepsilon_t^2)$.

2. Conditional on $\mathcal{I}_{t-1}$,

$$\mathbb{E}(y_t|\mathcal{I}_{t-1}) = \kappa + \theta y_{t-1}, \qquad \text{Var}(y_t|\mathcal{I}_{t-1}) = \alpha + \beta\varepsilon_{t-1}^2.$$

   Unconditionally,

$$\mathbb{E}(y_t) = \frac{\kappa}{1-\theta}, \qquad \text{Var}(\varepsilon_t) = \frac{\alpha}{1-\beta}, \qquad \text{Var}(y_t) = \frac{\alpha}{(1-\beta)(1-\theta^2)}.$$

3. The mean forecasts satisfy

$$\hat{y}_{T+h|T} = \frac{\kappa}{1-\theta} + \theta^h \left(y_T - \frac{\kappa}{1-\theta} \right).$$

The variance forecasts are

$$\hat{\sigma}^2_{T+1|T} = \alpha + \beta \varepsilon_T^2, \quad \hat{\sigma}^2_{T+2|T} = \alpha + \beta \hat{\sigma}^2_{T+1|T}, \quad \hat{\sigma}^2_{T+3|T} = \alpha + \beta \hat{\sigma}^2_{T+2|T}.$$

They converge to $\alpha/(1-\beta)$.

2.

1. Applying $1-(1-\lambda)L$ gives

$$y_t = \lambda\delta + (\lambda\beta_1 + \beta_2)x_t - (1-\lambda)\beta_2 x_{t-1} + (1-\lambda)y_{t-1} + \eta_t,$$

where $\eta_t = \varepsilon_t - (1-\lambda)\varepsilon_{t-1}$.

2. The impact multiplier is $\lambda\beta_1 + \beta_2$. For a one-period shock in x_t, the one-period interim multiplier is $\lambda(1-\lambda)\beta_1$. The long-run multiplier is $\beta_1 + \beta_2$; it is not changed by λ, although λ changes the speed.

3. Since the long-run relation is $y = \delta + (\beta_1 + \beta_2)x$,

$$\Delta y_t = (\lambda\beta_1 + \beta_2)\Delta x_t - \lambda\{y_{t-1} - \delta - (\beta_1 + \beta_2)x_{t-1}\} + \eta_t.$$

Exam 3 – Mock Exam 3

1. Univariate time series. Consider $\{\Delta y_t\}$ generated by

$$\Delta y_t = c + \beta \Delta y_{t-1} + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \gamma_2 \varepsilon_{t-2},$$

where $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ and $\beta, \gamma_1, \gamma_2 \neq 0$.

1. Assume $0 < \beta < 1$. Is $\{\Delta y_t\}$ stationary? What are the invertibility conditions? Can it have the same ACF as an $MA(1)$ process?
2. Assume $\beta = 1$. What are the orders of integration of $\{y_t\}$ and $\{\Delta y_t\}$?
3. Assume $0 < \beta < 1$ and $\gamma_2 = 0$. Find $\text{Var}(\Delta y_t)$. Given T observations, find forecasts for $\{y_t\}$ at $h = 1, 2$ and the associated forecast error variances.

2. Multivariate time series. Consider

$$y_t = c + \alpha y_{t-1} + \delta x_t + \gamma x_{t-1} + \varepsilon_t, \qquad x_t = k + \beta x_{t-1} + u_t + \rho u_{t-1},$$

with independent white noise processes ε_t and u_t.

1. Under what conditions on $\alpha, \beta, \gamma, \delta, \rho$ does $\mathbb{E}(y_t)$ exist? Find the relationship between $\mathbb{E}(y_t)$ and $\mathbb{E}(x_t)$ and discuss whether ρ matters.
2. Write the ADL model as an ECM for Δy_t.
3. Suppose $\alpha = \gamma = 0$ and $\beta = 1$. What are the orders of integration of $\{y_t\}$ and $\{x_t\}$? Are they cointegrated?

Solutions to Exam 3 – Mock Exam 3

1.

1. With $0 < \beta < 1$, the AR root is outside the unit circle, so Δy_t is stationary. Invertibility requires the roots of $1 + \gamma_1 z + \gamma_2 z^2 = 0$ to be outside the unit circle. The ACF is that of an $MA(1)$ if
$$1 + \gamma_1 L + \gamma_2 L^2 = (1 - \beta L)(1 + \delta L),$$
so $\gamma_1 = \delta - \beta$, $\gamma_2 = -\beta\delta$.

2. If $\beta = 1$ and no common factor cancels at $L = 1$, Δy_t is $I(1)$ and y_t is $I(2)$.

3. For $\gamma_2 = 0$,
$$\mathrm{Var}(\Delta y_t) = \frac{\sigma^2(1 + \gamma_1^2 + 2\beta\gamma_1)}{1 - \beta^2}.$$
Let $w_t = \Delta y_t$. Then
$$\hat{w}_{T+1|T} = c + \beta w_T + \gamma_1 \varepsilon_T, \qquad \hat{w}_{T+2|T} = c + \beta \hat{w}_{T+1|T},$$
$$\hat{y}_{T+1|T} = y_T + \hat{w}_{T+1|T}, \qquad \hat{y}_{T+2|T} = y_T + \hat{w}_{T+1|T} + \hat{w}_{T+2|T}.$$
The forecast error variances for levels are σ^2 and $\sigma^2\{1 + (1 + \beta + \gamma_1)^2\}$.

2.

1. $\mathbb{E}(y_t)$ exists if $|\alpha| < 1$ and $|\beta| < 1$; ρ does not affect the mean. Since $\mathbb{E}(x_t) = k/(1 - \beta)$,
$$\mathbb{E}(y_t) = \frac{c + (\delta + \gamma)\mathbb{E}(x_t)}{1 - \alpha}.$$

2.
$$\Delta y_t = \delta \Delta x_t - (1-\alpha)\left[y_{t-1} - \frac{c}{1-\alpha} - \frac{\delta+\gamma}{1-\alpha}x_{t-1}\right] + \varepsilon_t.$$
The correction speed is $1-\alpha$.

3. If $\alpha = \gamma = 0$ and $\beta = 1$, then x_t is $I(1)$ and $y_t = c + \delta x_t + \varepsilon_t$. If $\delta \neq 0$, y_t is $I(1)$ and $y_t - \delta x_t = c + \varepsilon_t$ is stationary; the series are cointegrated.

Exam 4 – Mock Exam 4

1. Univariate time series. Consider

$$\Delta y_t = k + \alpha_1 \Delta y_{t-1} + \alpha_2 \Delta y_{t-2} + \varepsilon_t + \delta_1 \varepsilon_{t-1}.$$

1. What are the stationarity conditions for $\{\Delta y_t\}$? Under those conditions, find $\mathbb{E}(\Delta y_t)$.
2. Can $\{\Delta y_t\}$ display the same ACF as $\Delta y_t = c + \rho \Delta y_{t-1} + \varepsilon_t$, with $\rho \neq 0$? If so, under what conditions?
3. Given T observations, find forecasts for $\{y_t\}$ at $h = 1, 2$ and the associated forecast error variances.

2. Multivariate time series. Consider

$$y_t = k + \rho y_{t-1} + \alpha x_t + \beta x_{t-1} + \eta_t, \qquad x_t = c + \varepsilon_t + \gamma \varepsilon_{t-1},$$

where η_t and ε_t are independent white noise processes.

1. Under what conditions do $\mathbb{E}(y_t)$ and $\mathrm{Var}(y_t)$ exist? Find them, or express the variance as a convergent filter variance.
2. Find the impact and one-period interim multipliers. Which multiplier is affected by ρ?
3. Find the long-run multiplier and write the ADL as an ECM for Δy_t.

Solutions to Exam 4 – Mock Exam 4

1.

1. For the AR(2) in Δy_t, stationarity requires the roots of $1-\alpha_1 z-\alpha_2 z^2=0$ to lie outside the unit circle; equivalently $\alpha_1+\alpha_2<1$, $\alpha_2-\alpha_1<1$, and $|\alpha_2|<1$. Then

$$\mathbb{E}(\Delta y_t)=\frac{k}{1-\alpha_1-\alpha_2}.$$

2. It reduces to an $AR(1)$ ACF if the MA factor cancels one AR factor:

$$1-\alpha_1 L-\alpha_2 L^2=(1-\rho L)(1+\delta_1 L).$$

   Hence $\alpha_1=\rho-\delta_1$, $\alpha_2=\rho\delta_1$, with $|\rho|<1$.

3. Let $w_t=\Delta y_t$:

$$\hat{w}_{T+1|T}=k+\alpha_1 w_T+\alpha_2 w_{T-1}+\delta_1\varepsilon_T,$$

$$\hat{w}_{T+2|T}=k+\alpha_1\hat{w}_{T+1|T}+\alpha_2 w_T.$$

   Thus $\hat{y}_{T+1|T}=y_T+\hat{w}_{T+1|T}$ and $\hat{y}_{T+2|T}=y_T+\hat{w}_{T+1|T}+\hat{w}_{T+2|T}$. The level forecast error variances are σ^2 and $\sigma^2\{1+(1+\alpha_1+\delta_1)^2\}$.

2.

1. $\mathbb{E}(y_t)$ and $\mathrm{Var}(y_t)$ exist if $|\rho|<1$; the MA process for x_t is stationary for finite γ. Since $\mathbb{E}(x_t)=c$,

$$\mathbb{E}(y_t)=\frac{k+(\alpha+\beta)c}{1-\rho}.$$

   For the variance, write the demeaned model as $(1-\rho L)\tilde{y}_t=(\alpha+\beta L)\tilde{x}_t+\eta_t$; this gives a convergent filter variance when $|\rho|<1$.

2. The impact multiplier is α. The one-period interim multiplier is $\rho\alpha+\beta$, so it is affected by ρ; the impact multiplier is not.

3. The long-run multiplier is $(\alpha+\beta)/(1-\rho)$ and

$$\Delta y_t=\alpha\Delta x_t-(1-\rho)\left[y_{t-1}-\frac{k}{1-\rho}-\frac{\alpha+\beta}{1-\rho}x_{t-1}\right]+\eta_t.$$

Exam 5 – Mock Exam 5

1. Univariate time series. Consider

$$\Delta y_t = c + \varepsilon_t + \gamma_1 \varepsilon_{t-1} + \gamma_2 \varepsilon_{t-2}.$$

1. Under what conditions is $\{\Delta y_t\}$ stationary? Is $\{y_t\}$ stationary? Under what conditions is $\{\Delta y_t\}$ invertible?

2. Suppose the process is invertible and estimate by OLS

$$\Delta y_t = \beta_0 + \beta_1 \Delta y_{t-1} + \beta_2 \Delta y_{t-2} + \beta_3 \Delta y_{t-3} + u_t.$$

   Can the estimated coefficients be interpreted as autocorrelations or partial autocorrelations? How does the answer change if $\gamma_2 = 0$?

3. Given T observations, find forecasts for $\{\Delta y_t\}$ at $h = 1, 2, 3$ and their forecast error variances. What happens as $h \to \infty$?

2. Multivariate time series. Consider

$$y_t = k + \alpha_1 x^*_{t+1} + \alpha_2 x_t + \varepsilon_t, \qquad x^*_{t+1} - x^*_t = \theta(x_t - x^*_t),$$

where $0 < \theta < 1$ and $\{x_t\}$ is stationary.

1. Derive an ADL model for y_t.

2. Find the impact, interim, and long-run multipliers. Does θ affect the long-run multiplier?

3. Construct the ECM for Δy_t.

Solutions to Exam 5 – Mock Exam 5

1.

1. Δy_t is an $MA(2)$, hence stationary for all finite γ_1, γ_2. The level y_t is generally $I(1)$. Invertibility requires the roots of $1 + \gamma_1 z + \gamma_2 z^2 = 0$ to be outside the unit circle.

2. In a regression with three lags, the coefficients are finite linear projection coefficients. The last coefficient corresponds to the partial autocorrelation at lag 3, while the earlier coefficients are not simple autocorrelations. If $\gamma_2 = 0$, the ACF cuts off after lag 1, but the PACF still tails off, so the same warning applies.

3. $$\hat{\Delta y}_{T+1|T} = c + \gamma_1 \varepsilon_T + \gamma_2 \varepsilon_{T-1}, \quad \hat{\Delta y}_{T+2|T} = c + \gamma_2 \varepsilon_T, \quad \hat{\Delta y}_{T+3|T} = c.$$
   The forecast error variances are σ^2, $\sigma^2(1 + \gamma_1^2)$, and $\sigma^2(1 + \gamma_1^2 + \gamma_2^2)$. The forecast converges to c and the variance to the unconditional variance.

2.

1. $$y_t = \theta k + (\theta\alpha_1 + \alpha_2)x_t - (1 - \theta)\alpha_2 x_{t-1} + (1 - \theta)y_{t-1} + \eta_t.$$

2. Impact: $\theta\alpha_1 + \alpha_2$. One-period interim: $\theta(1 - \theta)\alpha_1$. Long-run: $\alpha_1 + \alpha_2$, not affected by θ.

3. $$\Delta y_t = (\theta\alpha_1 + \alpha_2)\Delta x_t - \theta\{y_{t-1} - k - (\alpha_1 + \alpha_2)x_{t-1}\} + \eta_t.$$

Exam 6 – Mock Exam 6

1. Univariate time series. Consider

$$\Delta y_t = c + \beta \Delta y_{t-1} + \varepsilon_t + \gamma \varepsilon_{t-1}.$$

1. Under what conditions do the mean and variance of $\{\Delta y_t\}$ exist? What are the orders of integration of $\{\Delta y_t\}$ and $\{y_t\}$? What if $\beta = 1$?

2. Under stationarity, find $\mathbb{E}(\Delta y_t)$.

3. Given T observations, find forecasts for $\{y_t\}$ at $h = 1, 2$ and the associated forecast error variances.

2. Multivariate time series. Consider

$$y_t = \theta y_{t-1} + \beta x_t + \varepsilon_t, \qquad \varepsilon_t = u_t + \varphi u_{t-1}, \qquad x_t = \gamma x_{t-1} + v_t,$$

where u_t and v_t are independent white noise processes.

1. Under what conditions does $\mathbb{E}(y_t)$ exist? Find the relationship between $\mathbb{E}(y_t)$ and $\mathbb{E}(x_t)$.

2. Find the impact multiplier and the interim multipliers after one and two periods.

3. If $\theta = 0$ and $\gamma = 1$, are $\{y_t\}$ and $\{x_t\}$ cointegrated?

Solutions to Exam 6 – Mock Exam 6

1.

1. Mean and variance of Δy_t exist if $|\beta| < 1$; invertibility would additionally require $|\gamma| < 1$. Then Δy_t is $I(0)$ and y_t is $I(1)$. If $\beta = 1$ and there is no cancellation, Δy_t is $I(1)$ and y_t is $I(2)$.

2. $$\mathbb{E}(\Delta y_t) = \frac{c}{1-\beta}.$$

3. Let $w_t = \Delta y_t$:
$$\hat{w}_{T+1|T} = c + \beta w_T + \gamma \varepsilon_T, \qquad \hat{w}_{T+2|T} = c + \beta \hat{w}_{T+1|T}.$$
Then $\hat{y}_{T+1|T} = y_T + \hat{w}_{T+1|T}$ and $\hat{y}_{T+2|T} = y_T + \hat{w}_{T+1|T} + \hat{w}_{T+2|T}$. The forecast error variances are σ^2 and $\sigma^2\{1 + (1+\beta+\gamma)^2\}$.

2.

1. $\mathbb{E}(y_t)$ exists if $|\theta| < 1$ and $|\gamma| < 1$. With zero constants, $\mathbb{E}(x_t) = 0$ and $\mathbb{E}(y_t) = 0$; in general $\mathbb{E}(y) = \beta\mathbb{E}(x)/(1-\theta)$.

2. A unit innovation in v_t changes x_t, x_{t+1}, x_{t+2} by $1, \gamma, \gamma^2$. Thus the impact is β, and the interim multipliers are $\beta(\theta + \gamma)$ and $\beta(\theta^2 + \theta\gamma + \gamma^2)$.

3. If $\theta = 0$ and $\gamma = 1$, then x_t is $I(1)$ and $y_t = \beta x_t + \varepsilon_t$. For $\beta \neq 0$, $y_t - \beta x_t = \varepsilon_t$ is stationary, so the variables are cointegrated.

Exam 7 – Official Exam, 4 June 2026

1. Univariate time series. Consider

$$y_t = c + \theta_1 y_{t-1} + \theta_2 y_{t-2} + \varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2},$$

where c is a constant and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

1. Under what conditions do $\mathbb{E}(y_t)$ and $\text{Var}(y_t)$ exist? Under those conditions, find $\mathbb{E}(y_t)$.
2. Assume $\theta_1 = 2$ and $\theta_2 = -1$. Write the process for $\{\Delta y_t\}$. What are the orders of integration of $\{y_t\}$ and $\{\Delta y_t\}$?
3. Assume $\theta_1 = 1$, $\theta_2 = 0$, and $\alpha_2 = 0$. Given T observations, find forecasts for $\{y_t\}$ at $h = 1, 2$ and the associated forecast error variances. What happens as $h \to \infty$?

2. Multivariate time series. Consider

$$y_t^* = \beta x_t + \gamma y_{t-1} + u_t, \qquad y_t - y_{t-1} = \theta(y_t^* - y_{t-1}),$$

where $0 < \theta < 1$, $\{x_t\}$ is stationary with $\mathbb{E}(x_t) = \mu$, and $u_t \sim \text{WN}(0, \sigma^2)$.

1. Derive an ADL model for y_t.
2. Find the relationship between $\mathbb{E}(y_t)$ and μ and the associated long-run multiplier.
3. If the regression $y_t = \delta_0 + \delta_1 x_t + \delta_2 y_{t-1} + \varepsilon_t$ is estimated by OLS, is $\hat{\delta}_{1,OLS}$ consistent for β? Is it consistent for the impact multiplier?

Solutions to Exam 7 – Official Exam, 4 June 2026

1.

1. The roots of $1 - \theta_1 z - \theta_2 z^2 = 0$ must be outside the unit circle. Then
$$\mathbb{E}(y_t) = \frac{c}{1 - \theta_1 - \theta_2}.$$

2. With $\theta_1 = 2, \theta_2 = -1$,
$$(1 - L)^2 y_t = c + (1 + \alpha_1 L + \alpha_2 L^2)\varepsilon_t,$$
so
$$\Delta y_t = c + \Delta y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1} + \alpha_2 \varepsilon_{t-2}.$$
Without a common factor at $L = 1$, Δy_t is $I(1)$ and y_t is $I(2)$.

3. With $\theta_1 = 1, \theta_2 = 0, \alpha_2 = 0$,
$$y_t = c + y_{t-1} + \varepsilon_t + \alpha_1 \varepsilon_{t-1}.$$
Thus
$$\hat{y}_{T+1|T} = y_T + c + \alpha_1 \varepsilon_T, \qquad \hat{y}_{T+2|T} = y_T + 2c + \alpha_1 \varepsilon_T.$$
The forecast error variances are σ^2 and $\sigma^2\{1 + (1 + \alpha_1)^2\}$. As $h \to \infty$ the variance diverges.

2.

1.
$$y_t = \theta\beta x_t + [1 - \theta + \theta\gamma] y_{t-1} + \theta u_t.$$

2. Taking expectations,
$$\mathbb{E}(y_t) = \frac{\beta}{1 - \gamma}\mu,$$
so the long-run multiplier is $\beta/(1 - \gamma)$.

3. The population coefficient on x_t in the correctly specified ADL is $\theta\beta$. Therefore it is not consistent for β unless $\theta = 1$, but it is consistent for the impact multiplier $\theta\beta$ under the usual exogeneity assumptions.

Parte 2

Prove generate

I testi sono seguiti immediatamente dalle soluzioni della stessa prova.

Prove 8-20: esercitazioni generate.

Exam 8 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t + \eta \varepsilon_{t-1}.$$

1. State the stationarity and invertibility conditions. Under stationarity, find $\mathbb{E}(y_t)$.

2. Can this process have the same ACF as an $AR(1)$ process? Give the cancellation conditions.

3. Given T observations, find $\hat{y}_{T+1|T}$ and $\hat{y}_{T+2|T}$ and the forecast error variances.

2. Multivariate time series. Consider

$$y_t = a + \rho y_{t-1} + b_0 x_t + b_1 x_{t-1} + u_t,$$

where $\{x_t\}$ is stationary with mean μ_x.

1. Find conditions for $\mathbb{E}(y_t)$ to exist and derive $\mathbb{E}(y_t)$.

2. Find the impact, one-period interim, and long-run multipliers.

3. Write the ECM for Δy_t and interpret the adjustment coefficient.

Solutions to Exam 8 – Generated Practice Exam

1.

1. Stationarity requires the roots of $1 - \phi_1 z - \phi_2 z^2 = 0$ outside the unit circle; invertibility requires $|\eta| < 1$. Then $\mathbb{E}(y_t) = \mu/(1 - \phi_1 - \phi_2)$.

2. The process has an $AR(1)$ ACF if

$$1 - \phi_1 L - \phi_2 L^2 = (1 - rL)(1 + \eta L),$$

so $\phi_1 = r - \eta$, $\phi_2 = r\eta$, with $|r| < 1$.

3.

$$\hat{y}_{T+1|T} = \mu + \phi_1 y_T + \phi_2 y_{T-1} + \eta \varepsilon_T,$$
$$\hat{y}_{T+2|T} = \mu + \phi_1 \hat{y}_{T+1|T} + \phi_2 y_T.$$

The forecast error variances are σ^2 and $\sigma^2\{1 + (\phi_1 + \eta)^2\}$.

2.

1. $\mathbb{E}(y_t)$ exists if $|\rho| < 1$:
$$\mathbb{E}(y_t) = \frac{a + (b_0 + b_1)\mu_x}{1 - \rho}.$$
2. Impact $= b_0$, interim $= \rho b_0 + b_1$, long-run $= (b_0 + b_1)/(1 - \rho)$.
3.
$$\Delta y_t = b_0 \Delta x_t - (1 - \rho)\left[y_{t-1} - \frac{a}{1 - \rho} - \frac{b_0 + b_1}{1 - \rho} x_{t-1}\right] + u_t.$$

The adjustment coefficient is $-(1 - \rho)$.

Exam 9 – Generated Practice Exam

1. Univariate time series. Let

$$\Delta y_t = d + \varepsilon_t + \psi \varepsilon_{t-1}, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

1. Discuss stationarity of Δy_t, stationarity of y_t, and invertibility.
2. Find the ACF of Δy_t and explain how it identifies an $MA(1)$.
3. Find forecasts for y_{T+1}, y_{T+2}, and y_{T+3} and the forecast error variances.

2. Multivariate time series. Consider a partial-adjustment model

$$y_t^* = a + b x_t + g x_{t-1} + e_t, \qquad y_t - y_{t-1} = \lambda (y_t^* - y_{t-1}).$$

1. Derive the ADL representation.
2. Find the impact and long-run multipliers.
3. Explain how the speed of adjustment λ affects short-run and long-run effects.

Solutions to Exam 9 – Generated Practice Exam

1.

1. Δy_t is stationary for finite ψ; y_t is generally $I(1)$. Invertibility requires $|\psi| < 1$.
2. $\gamma_0 = \sigma^2(1+\psi^2)$, $\gamma_1 = \psi\sigma^2$, and $\gamma_j = 0$ for $j \geq 2$. Hence $\rho_1 = \psi/(1+\psi^2)$ and the ACF cuts off after lag 1.
3. $$\hat{y}_{T+1|T} = y_T + d + \psi\varepsilon_T, \quad \hat{y}_{T+2|T} = y_T + 2d + \psi\varepsilon_T, \quad \hat{y}_{T+3|T} = y_T + 3d + \psi\varepsilon_T.$$ The forecast error variances are σ^2, $\sigma^2\{1 + (1+\psi)^2\}$, and $\sigma^2\{1 + 2(1+\psi)^2\}$.

2.

1. $$y_t = \lambda a + \lambda b x_t + \lambda g x_{t-1} + (1-\lambda) y_{t-1} + \lambda e_t.$$
2. Impact $= \lambda b$; long-run $= (b+g)$.
3. λ scales the short-run response and the correction speed. The long-run effect is unchanged because the target is fully reached in equilibrium.

Exam 10 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = \kappa + \phi y_{t-1} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t v_t, \qquad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

with $v_t \sim \mathrm{WN}(0,1)$.

1. Give the conditions for weak stationarity and finite unconditional variance.
2. Find the conditional mean and variance of y_t and the unconditional mean and variance.
3. Forecast y_{T+h} and σ_{T+h}^2 for $h = 1, 2$ and describe the limit.

2. Multivariate time series. Consider

$$y_t = m + \delta x_{t+1}^* + \gamma x_t + \varepsilon_t, \qquad x_{t+1}^* - x_t^* = \lambda (x_t - x_t^*).$$

1. Derive the ADL model.
2. Find impact, interim, and long-run multipliers.
3. Derive the ECM and identify the equilibrium relation.

Solutions to Exam 10 – Generated Practice Exam

1.

1. Conditions: $|\phi| < 1$, $\omega > 0$, $\alpha, \beta \geq 0$, and $\alpha + \beta < 1$.

2.

$$\mathbb{E}(y_t|\mathcal{I}_{t-1}) = \kappa + \phi y_{t-1}, \qquad \text{Var}(y_t|\mathcal{I}_{t-1}) = \sigma_t^2.$$

$$\mathbb{E}(y_t) = \frac{\kappa}{1-\phi}, \qquad \text{Var}(\varepsilon_t) = \frac{\omega}{1-\alpha-\beta}, \qquad \text{Var}(y_t) = \frac{\omega}{(1-\alpha-\beta)(1-\phi^2)}.$$

3.

$$\hat{y}_{T+h|T} = \frac{\kappa}{1-\phi} + \phi^h \left(y_T - \frac{\kappa}{1-\phi} \right).$$

Also

$$\hat{\sigma}^2_{T+1|T} = \omega + \alpha\varepsilon_T^2 + \beta\sigma_T^2, \quad \hat{\sigma}^2_{T+2|T} = \omega + (\alpha+\beta)\hat{\sigma}^2_{T+1|T},$$

converging to $\omega/(1-\alpha-\beta)$.

2.

1.

$$y_t = \lambda m + (\lambda\delta + \gamma)x_t - (1-\lambda)\gamma x_{t-1} + (1-\lambda)y_{t-1} + \eta_t.$$

2. Impact $= \lambda\delta + \gamma$, interim $= \lambda(1-\lambda)\delta$, long-run $= \delta + \gamma$.

3.

$$\Delta y_t = (\lambda\delta + \gamma)\Delta x_t - \lambda[y_{t-1} - m - (\delta+\gamma)x_{t-1}] + \eta_t.$$

Exam 11 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = c + \rho y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

1. Discuss stationarity and invertibility.
2. If $\rho = 1$, derive the process for Δy_t and state the order of integration of y_t.
3. Under $|\rho| < 1$, find $\mathbb{E}(y_t)$ and the forecasts at $h = 1, 2$.

2. Multivariate time series. Let $x_t = x_{t-1} + v_t$ and

$$y_t = a + b x_t + u_t,$$

where u_t is stationary and independent of v_t.

1. What are the orders of integration of x_t and y_t?
2. Are x_t and y_t cointegrated? Give the cointegrating vector.
3. Write an ECM for Δy_t if u_t follows $u_t = \rho u_{t-1} + e_t$, $|\rho| < 1$.

Solutions to Exam 11 – Generated Practice Exam

1.

1. Stationarity requires $|\rho| < 1$; invertibility requires $|\theta| < 1$.
2. If $\rho = 1$,
$$\Delta y_t = c + \varepsilon_t + \theta \varepsilon_{t-1}.$$
Thus y_t is $I(1)$ unless the MA part cancels the unit root or the stochastic trend is absent.
3. Under $|\rho| < 1$,
$$\mathbb{E}(y_t) = \frac{c}{1-\rho}, \qquad \hat{y}_{T+1|T} = c + \rho y_T + \theta \varepsilon_T, \qquad \hat{y}_{T+2|T} = c + \rho \hat{y}_{T+1|T}.$$

2.

1. x_t is $I(1)$. If $b \neq 0$, y_t is also $I(1)$.
2. $y_t - a - bx_t = u_t$ is stationary, so the cointegrating vector is $(1, -b)$, with intercept $-a$.
3. Since $u_t = y_t - a - bx_t$ and $\Delta y_t = b\Delta x_t + \Delta u_t$,
$$\Delta y_t = b\Delta x_t - (1-\rho)(y_{t-1} - a - bx_{t-1}) + e_t.$$

Exam 12 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t.$$

1. State the AR(2) stationarity conditions and compute $\mathbb{E}(y_t)$.
2. Find the first two autocovariance restrictions implied by the Yule–Walker equations.
3. Given T observations, find forecasts at $h = 1, 2, 3$.

2. Multivariate time series. Consider the VAR(1)

$$\begin{pmatrix} y_t \\ x_t \end{pmatrix} = \begin{pmatrix} a_y \\ a_x \end{pmatrix} + A \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} + \begin{pmatrix} e_{yt} \\ e_{xt} \end{pmatrix}.$$

1. State the stationarity condition in terms of the eigenvalues of A.
2. Derive the unconditional mean vector.
3. Describe the impulse response at horizons $0, 1, 2$ to a shock in e_{xt}.

Solutions to Exam 12 – Generated Practice Exam

1.

1. Stationarity requires $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$, and $|\phi_2| < 1$. Then $\mathbb{E}(y_t) = c/(1 - \phi_1 - \phi_2)$.

2. For demeaned y_t, Yule–Walker gives

$$\gamma_1 = \phi_1\gamma_0 + \phi_2\gamma_1, \qquad \gamma_2 = \phi_1\gamma_1 + \phi_2\gamma_0.$$

3.

$$\hat{y}_{T+1|T} = c + \phi_1 y_T + \phi_2 y_{T-1},$$

$$\hat{y}_{T+2|T} = c + \phi_1 \hat{y}_{T+1|T} + \phi_2 y_T, \quad \hat{y}_{T+3|T} = c + \phi_1 \hat{y}_{T+2|T} + \phi_2 \hat{y}_{T+1|T}.$$

2.

1. The VAR is stationary if all eigenvalues of A have modulus less than one.

2.

$$\mathbb{E}(z_t) = (I - A)^{-1} a, \qquad z_t = (y_t, x_t)', \quad a = (a_y, a_x)'.$$

3. A unit shock to e_{xt} has impact vector $(0, 1)'$. The responses at horizons 1 and 2 are $A(0, 1)'$ and $A^2(0, 1)'$.

Exam 13 – Generated Practice Exam

1. Univariate time series. Let

$$\Delta y_t = c + \beta \Delta y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}, \qquad |\beta| < 1.$$

1. Give stationarity and invertibility conditions and find $\mathbb{E}(\Delta y_t)$.
2. Find $\text{Var}(\Delta y_t)$.
3. Find forecasts for y_{T+1} and y_{T+2} and their forecast error variances.

2. Multivariate time series. Consider

$$y_t = a + \rho y_{t-1} + b x_t + u_t, \qquad x_t = \mu + \gamma x_{t-1} + v_t.$$

1. State conditions for stationarity and find $\mathbb{E}(x_t)$ and $\mathbb{E}(y_t)$.
2. Compute the dynamic response of y_t, y_{t+1}, y_{t+2} to a unit shock v_t.
3. Find the long-run effect of a permanent change in x on y.

Solutions to Exam 13 – Generated Practice Exam

1.

1. Since $|\beta| < 1$, Δy_t is stationary; invertibility requires $|\theta| < 1$. Also $\mathbb{E}(\Delta y_t) = c/(1-\beta)$.

2. $$\operatorname{Var}(\Delta y_t) = \frac{\sigma^2(1 + \theta^2 + 2\beta\theta)}{1-\beta^2}.$$

3. $$\hat{\Delta y}_{T+1|T} = c + \beta \Delta y_T + \theta \varepsilon_T, \qquad \hat{\Delta y}_{T+2|T} = c + \beta \hat{\Delta y}_{T+1|T}.$$

   Then sum these forecasts to get level forecasts. The level forecast error variances are σ^2 and $\sigma^2\{1 + (1+\beta+\theta)^2\}$.

2.

1. Conditions: $|\rho| < 1$ and $|\gamma| < 1$. Then $\mathbb{E}(x_t) = \mu/(1-\gamma)$ and $\mathbb{E}(y_t) = \{a + b\mathbb{E}(x_t)\}/(1-\rho)$.

2. A unit v_t shock changes x_{t+j} by γ^j. Therefore the responses of y_t, y_{t+1}, y_{t+2} are

   $$b, \qquad b(\rho+\gamma), \qquad b(\rho^2 + \rho\gamma + \gamma^2).$$

3. The long-run effect of a permanent unit change in x is $b/(1-\rho)$.

Exam 14 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}.$$

1. Is $\{y_t\}$ stationary? State invertibility conditions.
2. Derive $\gamma_0, \gamma_1, \gamma_2$ and the ACF.
3. Find forecasts at $h = 1, 2, 3$ and forecast error variances.

2. Multivariate time series. Consider

$$y_t = \alpha + \beta x_t + \gamma z_t + u_t, \qquad z_t = \lambda z_{t-1} + (1-\lambda) x_t,$$

with $0 < \lambda < 1$ and stationary $\{x_t\}$.

1. Express y_t as an infinite distributed lag in x_t.
2. Find the impact and long-run multipliers of x on y.
3. Explain how λ affects the speed of adjustment.

Solutions to Exam 14 – Generated Practice Exam

1.

1. A finite MA process is stationary. Invertibility requires the roots of $1+\theta_1 z+\theta_2 z^2 = 0$ outside the unit circle.

2. $$\gamma_0 = \sigma^2(1+\theta_1^2+\theta_2^2), \quad \gamma_1 = \sigma^2(\theta_1+\theta_1\theta_2), \quad \gamma_2 = \sigma^2\theta_2,$$
   and $\gamma_j = 0$ for $j > 2$.

3.

$$\hat{y}_{T+1|T} = \mu + \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}, \quad \hat{y}_{T+2|T} = \mu + \theta_2 \varepsilon_T, \quad \hat{y}_{T+3|T} = \mu.$$

The forecast error variances are σ^2, $\sigma^2(1+\theta_1^2)$, and $\sigma^2(1+\theta_1^2+\theta_2^2)$.

2.

1. Since $z_t = (1-\lambda)\sum_{j=0}^{\infty} \lambda^j x_{t-j}$,

$$y_t = \alpha + \beta x_t + \gamma(1-\lambda)\sum_{j=0}^{\infty} \lambda^j x_{t-j} + u_t.$$

2. Impact $= \beta + \gamma(1-\lambda)$; long-run $= \beta + \gamma$.

3. Higher λ puts more weight on past x and slows adjustment, but the weights still sum to one.

Exam 15 – Generated Practice Exam

1. Univariate time series. Let $y_t = m_t + \eta_t$, where

$$m_t = m_{t-1} + \varepsilon_t,$$

and ε_t and η_t are independent white noise processes.

1. Is y_t stationary? What is the order of integration of y_t?
2. Show that Δy_t has an $MA(1)$ representation and give its autocovariances.
3. Find forecasts for y_{T+1} and y_{T+2}.

2. Multivariate time series. Consider

$$y_t = a + \rho y_{t-1} + b_0 x_t + b_1 x_{t-1} + e_t.$$

1. Find the long-run relation between y and x.
2. Derive the ECM for Δy_t.
3. Under what condition does the ECM imply convergence back to equilibrium?

Solutions to Exam 15 – Generated Practice Exam

1.

1. m_t is a random walk, so y_t is non-stationary and $I(1)$.

2. $$\Delta y_t = \varepsilon_t + \eta_t - \eta_{t-1}.$$
   Thus $\gamma_0 = \sigma_\varepsilon^2 + 2\sigma_\eta^2$, $\gamma_1 = -\sigma_\eta^2$, and $\gamma_j = 0$ for $j > 1$; it is observationally equivalent to an $MA(1)$.

3. If the components are observed or filtered at T, the best forecast is the current permanent component:
   $$\hat{y}_{T+1|T} = m_T, \qquad \hat{y}_{T+2|T} = m_T.$$
   The transitory shock has zero conditional mean.

2.

1. The long-run relation is
   $$y = \frac{a}{1-\rho} + \frac{b_0 + b_1}{1-\rho} x.$$

2. $$\Delta y_t = b_0 \Delta x_t - (1-\rho)\left[y_{t-1} - \frac{a}{1-\rho} - \frac{b_0 + b_1}{1-\rho} x_{t-1}\right] + e_t.$$

3. Convergence requires $0 < 1 - \rho < 2$ in the simple scalar adjustment interpretation; for standard stationary ADL dynamics, $|\rho| < 1$.

Exam 16 – Generated Practice Exam

1. Univariate time series. Consider

$$\Delta y_t = c + \alpha_1 \Delta y_{t-1} + \alpha_2 \Delta y_{t-2} + \varepsilon_t.$$

1. State the stationarity conditions for Δy_t.
2. If $\alpha_1 + \alpha_2 = 1$, discuss the order of integration of Δy_t and y_t.
3. Under stationarity, find forecasts for y_{T+1} and y_{T+2}.

2. Multivariate time series. Consider the VECM

$$\Delta z_t = \Pi z_{t-1} + e_t, \qquad z_t = (y_t, x_t)'.$$

1. Explain the implications of $\text{rank}(\Pi) = 0, 1, 2$.
2. If $\Pi = \alpha\beta'$ with rank one, identify the cointegrating relation and adjustment vector.
3. Explain what happens if the adjustment coefficient for y_t is zero.

Solutions to Exam 16 – Generated Practice Exam

1.

1. The AR(2) conditions are $\alpha_1 + \alpha_2 < 1$, $\alpha_2 - \alpha_1 < 1$, and $|\alpha_2| < 1$.
2. If $\alpha_1 + \alpha_2 = 1$, the AR polynomial for Δy_t has a unit root. Without cancellation, Δy_t is $I(1)$ and y_t is $I(2)$.
3. Under stationarity, let $w_t = \Delta y_t$:

$$\hat{w}_{T+1|T} = c + \alpha_1 w_T + \alpha_2 w_{T-1}, \qquad \hat{w}_{T+2|T} = c + \alpha_1 \hat{w}_{T+1|T} + \alpha_2 w_T.$$

   Then $\hat{y}_{T+1|T} = y_T + \hat{w}_{T+1|T}$ and $\hat{y}_{T+2|T} = y_T + \hat{w}_{T+1|T} + \hat{w}_{T+2|T}$.

2.

1. If $\text{rank}(\Pi) = 0$, there is no cointegration and the model is in differences. If the rank is one, there is one cointegrating relation. If the rank is two, z_t is stationary in levels.
2. With $\Pi = \alpha\beta'$, $\beta' z_t$ is the cointegrating error and α gives the adjustment speeds.
3. If the adjustment coefficient for y_t is zero, y_t does not respond directly to disequilibrium; adjustment occurs through the other variable.

Exam 17 – Generated Practice Exam

1. Univariate time series. Consider

$$r_t = \sigma_t v_t, \qquad \sigma_t^2 = \omega + \alpha r_{t-1}^2,$$

with $v_t \sim \mathrm{WN}(0,1)$.

1. Find conditions for positivity and covariance stationarity.
2. Find $\mathbb{E}(r_t)$, $\mathrm{Var}(r_t)$, and $\mathbb{E}(r_t^2 \mid \mathcal{I}_{t-1})$.
3. Forecast σ_{T+h}^2 for $h = 1, 2, 3$ and find the limit.

2. Multivariate time series. Consider

$$y_t = c + \beta x_t + \delta y_{t-1} + u_t, \qquad x_t = \mu + v_t.$$

1. Find $\mathbb{E}(y_t)$ and the long-run multiplier of x.
2. Find impact and interim multipliers.
3. Write the model in ECM form.

Solutions to Exam 17 – Generated Practice Exam

1.

1. Positivity requires $\omega > 0$ and $\alpha \geq 0$. Covariance stationarity requires $\alpha < 1$.
2. $\mathbb{E}(r_t) = 0$, $\mathrm{Var}(r_t) = \mathbb{E}(\sigma_t^2) = \omega/(1-\alpha)$, and $\mathbb{E}(r_t^2 | \mathcal{I}_{t-1}) = \sigma_t^2$.
3. $$\hat{\sigma}^2_{T+1|T} = \omega + \alpha r_T^2, \quad \hat{\sigma}^2_{T+2|T} = \omega + \alpha \hat{\sigma}^2_{T+1|T}, \quad \hat{\sigma}^2_{T+3|T} = \omega + \alpha \hat{\sigma}^2_{T+2|T},$$ with limit $\omega/(1-\alpha)$.

2.

1. If $|\delta| < 1$, $$\mathbb{E}(y_t) = \frac{c + \beta\mu}{1-\delta},$$ and the long-run multiplier is $\beta/(1-\delta)$.
2. Impact $= \beta$; one-period interim $= \delta\beta$ for a one-period change in x_t.
3. $$\Delta y_t = \beta \Delta x_t - (1-\delta)\left[y_{t-1} - \frac{c}{1-\delta} - \frac{\beta}{1-\delta} x_{t-1}\right] + u_t.$$

Exam 18 – Generated Practice Exam

1. Univariate time series. Consider the $ARIMA(1,1,1)$ model

$$\Delta y_t = c + \phi \Delta y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

1. Under what conditions is Δy_t stationary and invertible?
2. What is the order of integration of y_t?
3. Find forecasts for y_{T+1} and y_{T+2} and forecast error variances.

2. Multivariate time series. Suppose x_t is $I(1)$ and

$$y_t = a + b x_t + u_t, \qquad u_t = \rho u_{t-1} + e_t, \quad |\rho| < 1.$$

1. Show that y_t and x_t are cointegrated if $b \neq 0$.
2. Derive an ECM for Δy_t.
3. Is OLS in the cointegrating regression consistent for b?

Solutions to Exam 18 – Generated Practice Exam

1.

1. Δy_t is stationary if $|\phi| < 1$ and invertible if $|\theta| < 1$.
2. Then y_t is $I(1)$.
3. $$\widehat{\Delta y}_{T+1|T} = c + \phi \Delta y_T + \theta \varepsilon_T, \qquad \widehat{\Delta y}_{T+2|T} = c + \phi \widehat{\Delta y}_{T+1|T}.$$
   The level forecasts are obtained by summing the predicted differences. The level forecast error variances are σ^2 and $\sigma^2\{1 + (1 + \phi + \theta)^2\}$.

2.

1. Since $u_t = y_t - a - bx_t$ is stationary, y_t and x_t are cointegrated for $b \neq 0$.
2. $$\Delta y_t = b\Delta x_t - (1 - \rho)(y_{t-1} - a - bx_{t-1}) + e_t.$$
3. OLS in the cointegrating regression is consistent, in fact superconsistent under standard cointegration assumptions, although usual t inference is nonstandard.

Exam 19 – Generated Practice Exam

1. Univariate time series. Consider

$$y_t = c + \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

1. Under what condition does the ARMA process reduce to white noise plus a constant?
2. Under stationarity, find $\mathbb{E}(y_t)$ and the one-step forecast.
3. Find the long-run forecast and explain its dependence on ϕ.

2. Multivariate time series. Consider the bivariate system

$$y_t = a + \rho y_{t-1} + b x_{t-1} + e_t, \qquad x_t = \gamma x_{t-1} + v_t.$$

1. Give stationarity conditions.
2. Find $\mathbb{E}(y_t)$ and $\mathbb{E}(x_t)$.
3. Compute the response of y_{t+h} to a unit shock in v_t for $h = 1, 2, 3$.

Solutions to Exam 19 – Generated Practice Exam

1.

1. Since $(1 - \phi L)y_t = c + (1 + \theta L)\varepsilon_t$, the process reduces to white noise plus a constant if $1 + \theta L = 1 - \phi L$, i.e. $\theta = -\phi$.

2. Under $|\phi| < 1$, $\mathbb{E}(y_t) = c/(1 - \phi)$ and

$$\hat{y}_{T+1|T} = c + \phi y_T + \theta \varepsilon_T.$$

3.

$$\hat{y}_{T+h|T} \to \frac{c}{1 - \phi}$$

when $|\phi| < 1$. The speed is governed by ϕ^h.

2.

1. Stationarity requires $|\rho| < 1$ and $|\gamma| < 1$.

2. With no constants in x_t, $\mathbb{E}(x_t) = 0$ and $\mathbb{E}(y_t) = a/(1 - \rho)$.

3. A unit v_t shock affects x_t immediately, so y_{t+1} responds by b, y_{t+2} by $b(\rho + \gamma)$, and y_{t+3} by $b(\rho^2 + \rho\gamma + \gamma^2)$.

Exam 20 – Generated Practice Exam

1. Univariate time series. Consider

$$\Delta y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}.$$

1. Discuss stationarity of Δy_t, integration of y_t, and invertibility.
2. Find forecasts for Δy_{T+h} at $h = 1, 2, 3$.

3. Find forecasts for y_{T+h} at $h = 1, 2, 3$ and the associated forecast error variances.

2. Multivariate time series. Consider

$$y_t^* = a + bx_t + dy_{t-1} + u_t, \qquad y_t - y_{t-1} = \lambda(y_t^* - y_{t-1}).$$

1. Derive the ADL representation.
2. Find the impact and long-run multipliers.
3. Write the ECM and discuss how d and λ affect adjustment.

Solutions to Exam 20 – Generated Practice Exam

1.

1. Δy_t is stationary as an $MA(2)$; y_t is generally $I(1)$. Invertibility requires the roots of $1 + \theta_1 z + \theta_2 z^2 = 0$ outside the unit circle.

2. $$\widehat{\Delta y}_{T+1|T} = c + \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1}, \quad \widehat{\Delta y}_{T+2|T} = c + \theta_2 \varepsilon_T, \quad \widehat{\Delta y}_{T+3|T} = c.$$

3. Level forecasts are cumulative sums:

$$\hat{y}_{T+h|T} = y_T + \sum_{j=1}^{h} \widehat{\Delta y}_{T+j|T}.$$

   The forecast error variances are σ^2,

$$\sigma^2\{1 + (1+\theta_1)^2\},$$

   and

$$\sigma^2\{1 + (1+\theta_1)^2 + (1+\theta_1+\theta_2)^2\}.$$

2.

1. $$y_t = \lambda a + \lambda b x_t + [1 - \lambda + \lambda d] y_{t-1} + \lambda u_t.$$

2. Impact $= \lambda b$ and long-run multiplier $= b/(1-d)$.

3. $$\Delta y_t = \lambda b \Delta x_t - \lambda(1-d)\left[y_{t-1} - \frac{a}{1-d} - \frac{b}{1-d} x_{t-1}\right] + \lambda u_t.$$

   Higher λ speeds adjustment; higher d increases persistence and the long-run multiplier.

Altre tre prove generate

Testi e soluzioni spiegate

Approfondimenti su stima, consistenza, AIC/BIC e test DF-ADF-KPSS.

Econometria II – Tre prove d'esame generate

Time series: stima, consistenza, AIC/BIC, DF–ADF–KPSS

Testi d'esame e soluzioni spiegate con passaggi.

Struttura. Ogni prova contiene una parte univariata e una parte di interpretazione/diagnostica. Le soluzioni sono dopo i testi. La prima prova include un esercizio costruito sui temi del ripasso: stima AR/MA, ACF–PACF e consistenza.

Prova 1 – Estimation, ACF–PACF e consistency

Esercizio 1. Univariate time series. Sia

$$y_t = 0.8y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0,1).$$

1. Verificare se il processo è stazionario. Calcolare $\mathbb{E}(y_t)$, $\mathrm{Var}(y_t)$ e la funzione di autocorrelazione teorica ρ_k.
2. Dire quale comportamento ci si aspetta da ACF e PACF. Quale modello suggeriscono?
3. Si stimano 1000 simulazioni dello stesso processo per $T = 50, 100, 200, 1000$ e si calcolano le autocorrelazioni campionarie $\hat{\rho}_k$. Spiegare cosa significa dire che $\hat{\rho}_k$ è consistente.
4. Per l'AR(1), si stima ϕ con OLS. Sapendo che il bias di piccolo campione è circa

$$\mathbb{E}(\hat{\phi}_{OLS}) \simeq \phi\left(1 - \frac{2}{T}\right),$$

   calcolare il valore atteso approssimato di $\hat{\phi}_{OLS}$ per $T = 50$ e per $T = 200$. L'OLS è consistente?
5. Confrontare brevemente la stima di un $AR(p)$, di un $MA(1)$ invertibile e di un $ARMA(p,q)$.

Esercizio 2. Consistency in a partial-adjustment model. Si consideri

$$y_t^* = \alpha + \beta x_t + \gamma y_{t-1} + u_t, \qquad y_t - y_{t-1} = \lambda(y_t^* - y_{t-1}),$$

con $0 < \lambda < 1$, $|\gamma| < 1$, x_t stazionaria ed esogena rispetto a u_t, e $u_t \sim \mathrm{WN}(0, \sigma_u^2)$.

1. Derivare il modello ADL osservabile per y_t.
2. Se si stima con OLS

$$y_t = \delta_0 + \delta_1 x_t + \delta_2 y_{t-1} + e_t,$$

   per quali parametri strutturali sono consistenti $\hat{\delta}_0, \hat{\delta}_1, \hat{\delta}_2$?
3. $\hat{\delta}_1$ è consistente per β? È consistente per l'impact multiplier?
4. Calcolare il long-run multiplier di x su y e spiegare perché non coincide con l'impact multiplier.

Domande brevi.

1. In un $AR(p)$, quale funzione si azzera dopo il lag p: ACF o PACF?
2. In un $MA(q)$ invertibile, l'OLS diretto sui lag di y_t stima direttamente i parametri MA?
3. Un bias di piccolo campione diverso da zero implica necessariamente inconsistenza?

Prova 2 – AIC, BIC, Ljung–Box e forecast ARMA

Esercizio 1. Model selection. Su un campione di ampiezza $T = 80$ sono stati stimati tre modelli ARMA con costante. La varianza residua stimata è indicata nella tabella:

Modello	(p, q)	$k = p + q + 1$	$\hat{\sigma}^2$
M_1	$(1, 1)$	3	1.00
M_2	$(2, 2)$	5	0.94
M_3	$(2, 0)$	3	1.08

Usare

$$\mathrm{AIC} = \log(\hat{\sigma}^2) + \frac{2k}{T}, \qquad \mathrm{BIC} = \log(\hat{\sigma}^2) + \frac{k \log T}{T}.$$

1. Calcolare AIC e BIC per i tre modelli.
2. Quale modello sceglie AIC? Quale sceglie BIC?
3. Spiegare perché AIC e BIC possono scegliere modelli diversi.
4. Per M_2 si ottiene un Ljung–Box $Q_{12} = 15.5$. Usando la regola $Q_K \sim \chi^2_{K-p-q-1}$, valutare se i residui sono white noise al 5%, sapendo che $\chi^2_{7,0.95} = 14.07$.

Esercizio 2. Forecast di un ARMA(1,1). Sia

$$y_t = c + \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2),$$

con $c = 0.2$, $\phi = 0.6$, $\theta = 0.4$, $\sigma^2 = 1.2$. Al tempo T si osservano $y_T = 1.5$ ed $\varepsilon_T = 0.3$.

1. Verificare stazionarietà e invertibilità.
2. Calcolare $\hat{y}_{T+1|T}$, $\hat{y}_{T+2|T}$ e $\hat{y}_{T+3|T}$.
3. Calcolare il limite del forecast per $h \to \infty$.
4. Calcolare le forecast error variances per $h = 1$ e $h = 2$.

Domande brevi.

1. Se un modello ha AIC più basso ma residui autocorrelati, va scelto automaticamente?
2. BIC penalizza di più o di meno il numero di parametri rispetto ad AIC per campioni ragionevoli?
3. Perché un $ARMA(2, 2)$ può ridursi a un modello più semplice?

Prova 3 – DF, ADF, KPSS e unit root

Esercizio 1. Test di radice unitaria. Si consideri una serie y_t e la regressione Dickey–Fuller con costante

$$\Delta y_t = a + \gamma y_{t-1} + u_t.$$

1. Scrivere ipotesi nulla e alternativa del test DF in termini di γ e in termini di ordine di integrazione.
2. Spiegare come cambia la regressione nel test ADF e perché si aggiungono lag di Δy_t.
3. Spiegare la differenza fondamentale tra DF/ADF e KPSS in termini di ipotesi nulla.

Esercizio 2. Interpretazione di risultati empirici. Si hanno i seguenti risultati:

Serie	Test	Statistica	Valore critico 5%
A	DF con costante	-2.30	-2.86
A	$KPSS$ livello	0.62	0.463
B	ADF con costante e trend	-4.10	-3.41
B	$KPSS$ trend	0.09	0.146
C	ADF su z_t con costante	-1.80	-2.86
C	ADF su Δz_t con costante	-5.20	-2.86

1. Interpretare la serie A combinando DF e KPSS.
2. Interpretare la serie B combinando ADF e KPSS.
3. Stabilire l'ordine di integrazione più plausibile di z_t.

Esercizio 3. Radice unitaria da polinomio AR. Sia

$$(1 - 1.3L + 0.3L^2)y_t = \varepsilon_t, \qquad \varepsilon_t \sim \mathrm{WN}(0, \sigma^2).$$

1. Fattorizzare il polinomio autoregressivo.
2. Stabilire se y_t è stazionaria oppure ha una unit root.
3. Scrivere il processo seguito da Δy_t.
4. Dire quali risultati ci si aspetta da un ADF su y_t e da un ADF su Δy_t.

Soluzioni – Prova 1

Esercizio 1

1. Il processo è un $AR(1)$ con $\phi = 0.8$. La condizione di stazionarietà è $|\phi| < 1$, quindi è soddisfatta. Poiché non c'è costante,
$$\mathbb{E}(y_t) = 0.$$
Per la varianza:
$$\mathrm{Var}(y_t) = \mathrm{Var}(0.8y_{t-1} + \varepsilon_t) = 0.8^2 \,\mathrm{Var}(y_{t-1}) + 1.$$
In stazionarietà $\mathrm{Var}(y_t) = \mathrm{Var}(y_{t-1}) = \gamma_0$, quindi
$$\gamma_0 = 0.64\gamma_0 + 1 \quad \Rightarrow \quad (1 - 0.64)\gamma_0 = 1 \quad \Rightarrow \quad \gamma_0 = \frac{1}{0.36} = \frac{25}{9}.$$
Per un $AR(1)$ stazionario,
$$\gamma_k = \phi^k \gamma_0, \qquad \rho_k = \frac{\gamma_k}{\gamma_0} = \phi^k.$$
Quindi
$$\rho_k = 0.8^k.$$

2. L'ACF decresce gradualmente in modo geometrico: $0.8, 0.8^2, 0.8^3, \ldots$. La PACF invece si annulla dopo il primo lag. Questo è il pattern tipico di un $AR(1)$: ACF tailing off, PACF cut-off dopo $k = 1$.

3. Dire che $\hat{\rho}_k$ è consistente significa che, fissato il lag k,
$$\mathrm{plim}_{T\to\infty} \hat{\rho}_k = \rho_k.$$
Nel caso specifico,
$$\mathrm{plim}\, \hat{\rho}_k = 0.8^k.$$
Intuitivamente: con pochi dati le autocorrelazioni campionarie oscillano molto; aumentando T, la loro distribuzione si concentra intorno al valore vero. Questa proprietà richiede stazionarietà ed ergodicità: il campione lungo deve rappresentare bene il processo.

4. Per $T = 50$:
$$\mathbb{E}(\hat{\phi}_{OLS}) \simeq 0.8\left(1 - \frac{2}{50}\right) = 0.8(0.96) = 0.768.$$
Per $T = 200$:
$$\mathbb{E}(\hat{\phi}_{OLS}) \simeq 0.8\left(1 - \frac{2}{200}\right) = 0.8(0.99) = 0.792.$$
Il bias è negativo e diminuisce quando T aumenta. Infatti
$$\lim_{T\to\infty} 0.8\left(1 - \frac{2}{T}\right) = 0.8.$$
Quindi l'OLS può essere biased in finite sample ma resta consistente: bias di piccolo campione non implica inconsistenza.

5. Per un $AR(p)$ si può usare OLS perché il modello è lineare nei parametri autoregressivi:
$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t.$$
Per un $MA(1)$,
$$y_t = \varepsilon_t + \theta\varepsilon_{t-1},$$
l'errore passato ε_{t-1} non è osservato. Se il processo è invertibile, si può esprimere l'errore come funzione dei passati y, ma la stima è non lineare: si usano NLS, conditional least squares o maximum likelihood. Un $ARMA(p, q)$ combina le due difficoltà: la parte AR è lineare, ma la parte MA richiede una procedura non lineare.

Esercizio 2

1. Partiamo da
$$y_t - y_{t-1} = \lambda(y_t^* - y_{t-1}).$$
Sostituiamo $y_t^* = \alpha + \beta x_t + \gamma y_{t-1} + u_t$:
$$y_t - y_{t-1} = \lambda\left(\alpha + \beta x_t + \gamma y_{t-1} + u_t - y_{t-1}\right).$$
Quindi
$$y_t - y_{t-1} = \lambda\alpha + \lambda\beta x_t + \lambda(\gamma - 1)y_{t-1} + \lambda u_t.$$
Portando y_{t-1} a destra:
$$y_t = \lambda\alpha + \lambda\beta x_t + \left[1 + \lambda(\gamma - 1)\right] y_{t-1} + \lambda u_t.$$
Poiché $1 + \lambda(\gamma - 1) = 1 - \lambda(1 - \gamma)$,
$$\boxed{y_t = \lambda\alpha + \lambda\beta x_t + \left[1 - \lambda(1 - \gamma)\right] y_{t-1} + \lambda u_t.}$$

2. Confrontando con
$$y_t = \delta_0 + \delta_1 x_t + \delta_2 y_{t-1} + e_t,$$
si ottiene
$$\delta_0 = \lambda\alpha, \qquad \delta_1 = \lambda\beta, \qquad \delta_2 = 1 - \lambda(1 - \gamma), \qquad e_t = \lambda u_t.$$
Se x_t è esogena e u_t non è correlato con i regressori, OLS è consistente per i parametri della regressione osservabile, cioè per $\delta_0, \delta_1, \delta_2$.

3. $\hat{\delta}_1$ è consistente per $\lambda\beta$, non per β, a meno che $\lambda = 1$ oppure λ sia noto e si divida $\hat{\delta}_1$ per λ. L'impact multiplier, cioè l'effetto contemporaneo di x_t su y_t, è
$$\frac{\partial y_t}{\partial x_t} = \lambda\beta.$$
Quindi $\hat{\delta}_1$ è consistente per l'impact multiplier.

4. Il long-run multiplier si calcola imponendo equilibrio: $y_t = y_{t-1} = y$ e $x_t = x$. Dal modello osservabile:
$$y = \lambda\alpha + \lambda\beta x + \left[1 - \lambda(1 - \gamma)\right] y.$$
Portiamo il termine in y a sinistra:
$$y\left(1 - 1 + \lambda(1 - \gamma)\right) = \lambda\alpha + \lambda\beta x.$$
Quindi
$$\lambda(1 - \gamma)y = \lambda\alpha + \lambda\beta x.$$
Dividendo per λ:
$$(1 - \gamma)y = \alpha + \beta x.$$
Pertanto
$$y = \frac{\alpha}{1 - \gamma} + \frac{\beta}{1 - \gamma}x, \qquad \boxed{LRM = \frac{\beta}{1 - \gamma}.}$$
L'impact multiplier è $\lambda\beta$, mentre il long-run multiplier è $\beta/(1 - \gamma)$. Il primo misura l'effetto immediato; il secondo incorpora tutti gli aggiustamenti futuri dovuti alla persistenza di y_{t-1}.

Domande brevi

1. Nell'$AR(p)$ si azzera la PACF dopo il lag p; l'ACF decresce gradualmente.

2. No. In un $MA(q)$ gli errori passati non sono osservati; i parametri MA richiedono stima non lineare, sotto invertibilità.

3. No. Un bias che va a zero quando $T \to \infty$ è compatibile con la consistenza.

Soluzioni – Prova 2

Esercizio 1

Usiamo $\log(80) = 4.3820$.

$$\text{AIC} = \log(\hat{\sigma}^2) + \frac{2k}{80}, \qquad \text{BIC} = \log(\hat{\sigma}^2) + \frac{k \log 80}{80}.$$

Modello	$\log(\hat{\sigma}^2)$	k	AIC	BIC
M_1	$\log(1.00) = 0$	3	$0 + 6/80 = 0.0750$	$0 + 3(4.3820)/80 = 0.1643$
M_2	$\log(0.94) = -0.0619$	5	$-0.0619 + 10/80 = 0.0631$	$-0.0619 + 5(4.3820)/80 = 0.2120$
M_3	$\log(1.08) = 0.0770$	3	$0.0770 + 6/80 = 0.1520$	$0.0770 + 3(4.3820)/80 = 0.2413$

1. I valori sono quelli riportati in tabella.

2. AIC sceglie il modello con valore più basso: M_2, perché 0.0631 è il minimo. BIC sceglie M_1, perché 0.1643 è il minimo.

3. AIC e BIC premiano entrambi una bassa varianza residua, ma penalizzano il numero di parametri. La penalizzazione BIC è $k \log(T)/T$, mentre quella AIC è $2k/T$. Poiché per $T = 80$ vale $\log(T) > 2$, BIC penalizza di più la complessità. Per questo AIC può scegliere un modello più ricco e BIC uno più parsimonioso.

4. Per M_2, $p = 2$ e $q = 2$. Con la regola data,

   $$df = K - p - q - 1 = 12 - 2 - 2 - 1 = 7.$$

   La statistica è $Q_{12} = 15.5$. Poiché

   $$15.5 > \chi^2_{7,0.95} = 14.07,$$

   rifiutiamo l'ipotesi nulla di assenza di autocorrelazione residua fino al lag 12. Quindi i residui non sembrano white noise: anche se M_2 ha AIC basso, il diagnostico segnala possibile misspecificazione.

Esercizio 2

1. La stazionarietà dipende dalla parte AR: serve $|\phi| < 1$. Qui $|0.6| < 1$, quindi il processo è stazionario. L'invertibilità di un $MA(1)$ richiede $|\theta| < 1$. Qui $|0.4| < 1$, quindi il processo è invertibile.

2. Al tempo T, $\mathbb{E}_T(\varepsilon_{T+h}) = 0$ per $h \geq 1$. Per $h = 1$:

   $$\hat{y}_{T+1|T} = 0.2 + 0.6 y_T + 0.4 \varepsilon_T = 0.2 + 0.6(1.5) + 0.4(0.3) = 1.22.$$

   Per $h = 2$:

   $$\hat{y}_{T+2|T} = 0.2 + 0.6 \hat{y}_{T+1|T} = 0.2 + 0.6(1.22) = 0.932.$$

   Per $h = 3$:

   $$\hat{y}_{T+3|T} = 0.2 + 0.6 \hat{y}_{T+2|T} = 0.2 + 0.6(0.932) = 0.7592.$$

3. Il forecast di lungo periodo converge alla media incondizionata:

   $$\mu = \frac{c}{1 - \phi} = \frac{0.2}{1 - 0.6} = 0.5.$$

   Quindi

   $$\hat{y}_{T+h|T} \to 0.5.$$

4. Scriviamo la rappresentazione $MA(\infty)$. Per un $ARMA(1,1)$,

$$y_t - \mu = \varepsilon_t + (\phi + \theta)\varepsilon_{t-1} + \phi(\phi + \theta)\varepsilon_{t-2} + \cdots .$$

Qui

$$\psi_0 = 1, \qquad \psi_1 = \phi + \theta = 0.6 + 0.4 = 1.$$

La forecast error variance a un passo è

$$\mathrm{Var}(y_{T+1} - \hat{y}_{T+1|T}) = \sigma^2 = 1.2.$$

A due passi:

$$\mathrm{Var}(y_{T+2} - \hat{y}_{T+2|T}) = \sigma^2(1 + \psi_1^2) = 1.2(1 + 1^2) = 2.4.$$

Domande brevi

1. No. Prima bisogna controllare i residui: un modello con AIC basso ma residui autocorrelati può essere misspecificato.
2. BIC penalizza di più, perché per campioni ragionevoli $\log(T) > 2$.
3. Perché i polinomi AR e MA possono avere un fattore comune. Se il fattore si cancella, un $ARMA(2,2)$ può avere la stessa struttura di autocovarianza di un modello di ordine inferiore.

Soluzioni – Prova 3

Esercizio 1

1. Il modello DF con costante è

$$\Delta y_t = a + \gamma y_{t-1} + u_t.$$

Deriva da un $AR(1)$:

$$y_t = a + \theta y_{t-1} + u_t.$$

Sottraendo y_{t-1} da entrambi i lati:

$$\Delta y_t = a + (\theta - 1) y_{t-1} + u_t.$$

Quindi

$$\gamma = \theta - 1.$$

Le ipotesi sono

$$H_0 : \gamma = 0 \quad \Leftrightarrow \quad \theta = 1 \quad \Rightarrow \quad y_t \sim I(1),$$
$$H_1 : \gamma < 0 \quad \Leftrightarrow \quad |\theta| < 1 \quad \Rightarrow \quad y_t \sim I(0)$$

nel caso senza radici esplosive e con specificazione deterministica corretta.

2. Il test ADF aggiunge lag di Δy_t:

$$\Delta y_t = a + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + u_t.$$

Con trend:

$$\Delta y_t = a + bt + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + u_t.$$

I lag di Δy_t servono a eliminare autocorrelazione nei residui. L'idea resta la stessa: si testa $\gamma = 0$ contro $\gamma < 0$.

3. DF e ADF hanno come nulla la presenza di unit root:

$$H_0 : y_t \sim I(1).$$

KPSS invece ha nulla opposta:

$$H_0 : y_t \text{ è stazionaria}$$

oppure, nella versione con trend,

$$H_0 : y_t \text{ è trend-stationary.}$$

Per questo DF/ADF e KPSS sono complementari: se ADF non rifiuta unit root e KPSS rifiuta stazionarietà, l'evidenza a favore di $I(1)$ è più forte.

Esercizio 2

1. Per la serie A, il DF con costante dà -2.30. Nel test DF si rifiuta se la statistica è più negativa del valore critico. Poiché

$$-2.30 > -2.86,$$

non rifiutiamo H_0 di unit root. Il KPSS livello dà 0.62 e si rifiuta la stazionarietà se la statistica supera il critico:

$$0.62 > 0.463.$$

Quindi KPSS rifiuta la stazionarietà. Combinando i due risultati: la serie A è plausibilmente $I(1)$.

2. Per la serie B, l'ADF con costante e trend dà -4.10. Poiché

$$-4.10 < -3.41,$$

rifiutiamo la unit root. Il KPSS trend dà 0.09 e

$$0.09 < 0.146,$$

quindi non rifiutiamo la trend-stationarity. Conclusione: B è plausibilmente stazionaria attorno a un trend deterministico, non $I(1)$.

3. Per z_t, l'ADF sul livello dà -1.80, quindi

$$-1.80 > -2.86,$$

non rifiutiamo unit root nel livello. L'ADF sulla differenza dà -5.20, quindi

$$-5.20 < -2.86,$$

rifiutiamo unit root in Δz_t. Dunque Δz_t è plausibilmente $I(0)$ e

$$\boxed{z_t \sim I(1).}$$

Esercizio 3

1. Il polinomio è

$$1 - 1.3L + 0.3L^2.$$

Cerchiamo la fattorizzazione:

$$(1 - L)(1 - 0.3L) = 1 - 0.3L - L + 0.3L^2 = 1 - 1.3L + 0.3L^2.$$

Quindi

$$\boxed{1 - 1.3L + 0.3L^2 = (1 - L)(1 - 0.3L).}$$

2. Il fattore $(1 - L)$ indica una unit root. Quindi y_t non è stazionaria in livello. Poiché l'altro fattore ha coefficiente 0.3 con $|0.3| < 1$, dopo una differenza resta una parte AR stazionaria.

3. Partiamo da

$$(1 - L)(1 - 0.3L)y_t = \varepsilon_t.$$

Poiché $(1 - L)y_t = \Delta y_t$,

$$(1 - 0.3L)\Delta y_t = \varepsilon_t.$$

Quindi

$$\boxed{\Delta y_t = 0.3\Delta y_{t-1} + \varepsilon_t.}$$

La differenza prima segue un $AR(1)$ stazionario.

4. Su y_t ci aspettiamo che un ADF non rifiuti la null di unit root, perché y_t contiene il fattore $(1 - L)$. Su Δy_t ci aspettiamo invece che un ADF rifiuti la null di unit root, perché Δy_t è un $AR(1)$ stazionario con coefficiente 0.3.